

Automated Product Image Collection and Training for Object Detection

Week 2

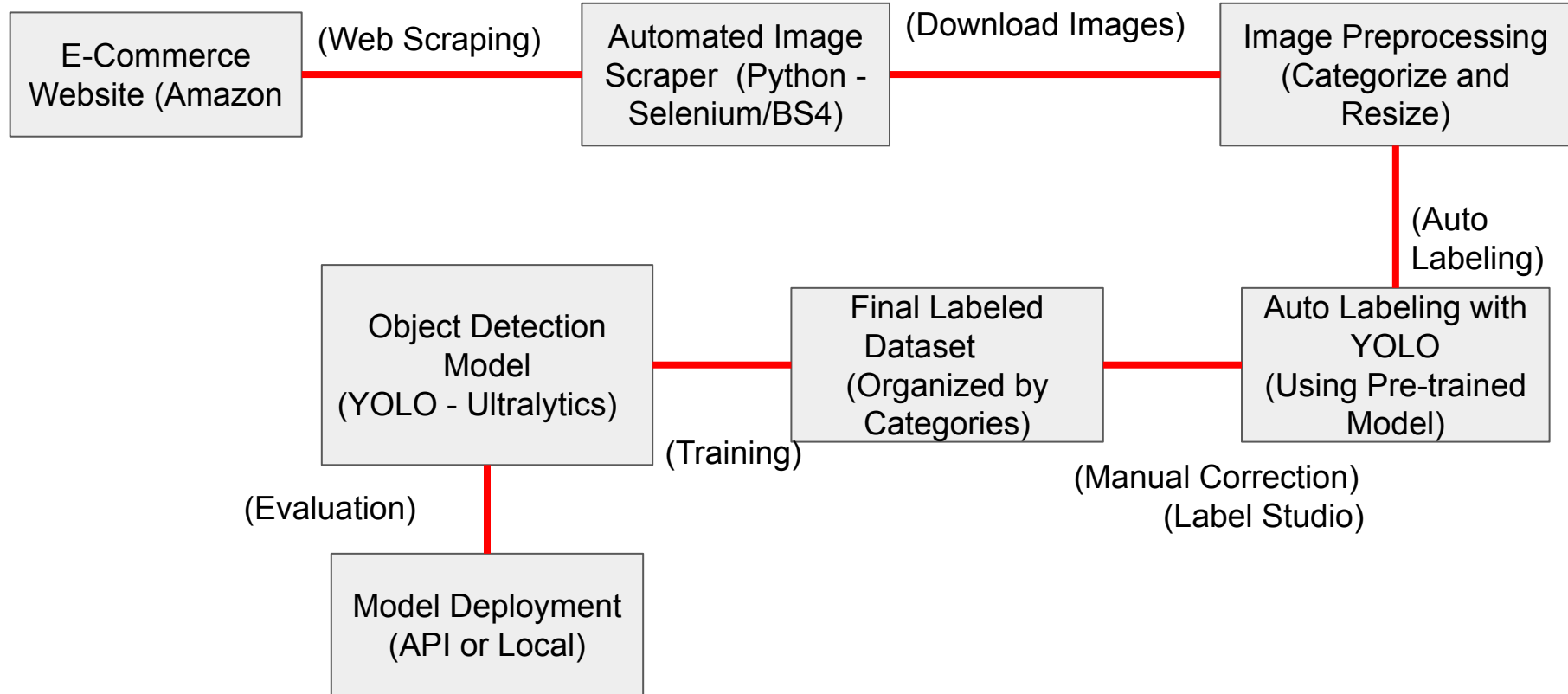
1. Objective:

Develop an automated pipeline for collecting product images from e-commerce websites (e.g., Amazon) and training a robust object detection model using these images.

[Amazon.com: Moschino, Chambred Handle Leather Bag, Red : Luxury Stores](#)



2. System Architecture:



3. Detailed Steps:

Step 1: Automated Image Collection

- Use `Selenium` or `Beautiful Soup` in Python to scrape images from Amazon or other e-commerce websites.
- Collect images for multiple product categories (e.g., Electronics, Fashion, Groceries).
- Save images in a structured folder format

Step 2: Image Preprocessing

- Resize images for consistency (e.g., 640x640).
- Apply data augmentation techniques:
 - Flip (horizontal/vertical)
 - Rotate (0-360 degrees)
 - Brightness adjustment
 - Contrast enhancement

Step 3: Automated Labeling

- Use a pre-trained YOLO model for initial object detection and labeling.
- Run the pre-trained model on the collected images to generate bounding boxes.
- Store these initial labels in COCO format (JSON).

Step 4: Manual Labeling Correction

- Use Label Studio to review and correct the automatically generated labels.
- Adjust bounding boxes for accuracy.
- Save the corrected labeled dataset.

Step 5: Training the YOLO Model

- Set up YOLO (Ultralytics) environment:
 - Use the collected and labeled dataset.
 - Split the dataset into training (70%), validation (20%), and test (10%)
- Configure the YOLO training parameters:
 - Learning rate
 - Batch size
 - Number of epochs (e.g., 50-100)
- Train the model.

Step 6: Model Evaluation

- Test the model on unseen images.
- Evaluate using metrics like:
 - Precision
 - Recall
 - mAP (Mean Average Precision)

Step 7: Deployment

- Deploy the model using:
 - A local application (Python GUI)
 - A cloud API (FastAPI, Flask)
 - Integrate with a camera for real-time detection.